

Bio Informatics

Lecture 10

Codon Table

		Second letter					
		U	C	A	G		
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U	C
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U	C
	A	AUU } Ile AUC } AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U	C
	G	GUU } Val GUC } GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U	C
						A	G
						Third letter	

Review

The Central Dogma of Molecular Biology

DNA

Review

The Central Dogma of Molecular Biology

DNA → mRNA

Review

The Central Dogma of Molecular Biology



CTC

Review

The Central Dogma of Molecular Biology

DNA → **mRNA** → **Proteins**

CTC → **GAG**

Review

The Central Dogma of Molecular Biology

DNA → **mRNA** → **Proteins**

CTC → **GAG** → **Glu**

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The Central Dogma of Molecular Biology

DNA → **mRNA** → **Proteins**

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CAC

Review

The Central Dogma of Molecular Biology

DNA → **mRNA** → **Proteins**

CTC → **GAG** → **Glu**

CAC → **GUG**

Review

The Central Dogma of Molecular Biology

DNA → **mRNA** → **Proteins**

CTC → **GAG** → **Glu**

CAC → **GUG** → **Val**

Genetic Mutation

- What is a mutation?
- A mutation is a change that occurs in our DNA sequence, either due to mistakes when the DNA is copied or as the result of environmental factors such as UV light (google: Thymine dimer) and cigarette smoke.

Genetic Mutation

- Over a lifetime our DNA can undergo changes or 'mutations' in the sequence of bases A, C, G and T.
- This results in changes in the proteins that are made. This can be a bad or a good thing.

Genetic Mutation

- Mutations can occur during DNA replication if errors are made and not corrected in time.
- Mutations can also occur as the result of exposure to environmental factors such as smoking, sunlight and radiation.
- Often cells can recognize any potentially mutation-causing damage and repair it before it becomes a fixed mutation.

Genetic Mutation

- Mutations contribute to genetic variation within species.
- Mutations can also be inherited, whether they have a positive or a negative effect.
- Some well-known **inherited genetic** disorders include cystic fibrosis, sickle cell anemia and color-blindness, among many others. All of these disorders are caused by the **mutation** of a single gene.

Genetic Mutation

- For example, the disorder sickle cell anemia is caused by a mutation in the gene that instructs the building of a protein called hemoglobin. This causes the red blood cells to become an abnormal, rigid, sickle shape.
- On the contrary, individuals who are carriers for the **sickle cell disease** have some protective advantage against **malaria**.

Genetic Mutation

- However, mutation can also disrupt normal gene activity and cause diseases, like cancer.
- Cancer is the most common human genetic disease; it is caused by mutations occurring in a number of growth-controlling genes.
- Sometimes faulty, cancer-causing genes can exist from birth, increasing a person's chance of getting cancer.

Types of Genetic Mutations

- **Point mutation**
- **Substitution mutation**
- **Inversion mutation**
- **Frame-Shift mutation**

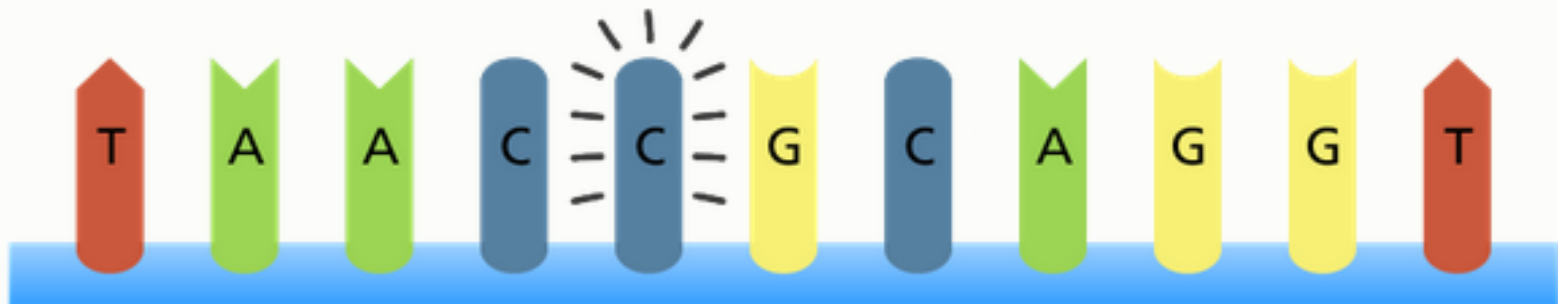
Point Mutation

- **Point mutation** – a change in one base in the DNA sequence.

Original sequence



Point mutation



Point Mutation – Silent

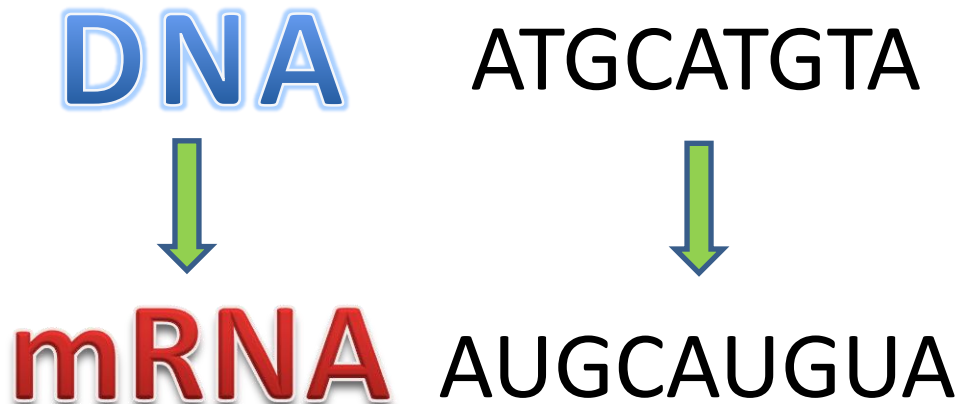
- **Point (Silent) mutation** – are **mutations** in DNA that do not have an observable effect on the organism's phenotype. They are a specific type of neutral **mutation**

DNA

ATGCATGTA

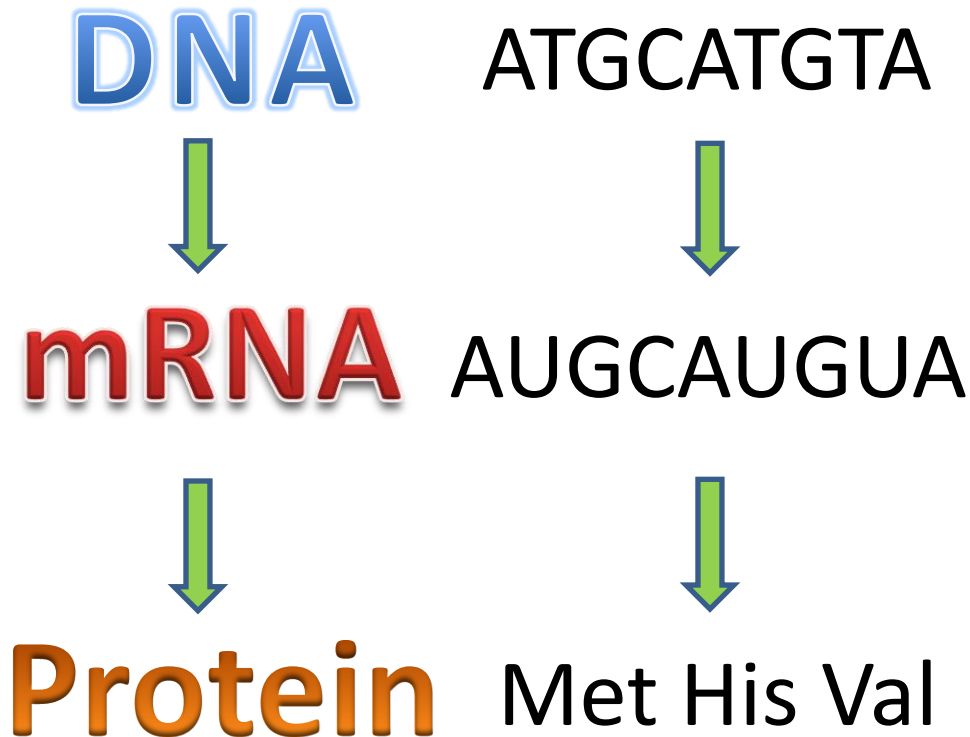
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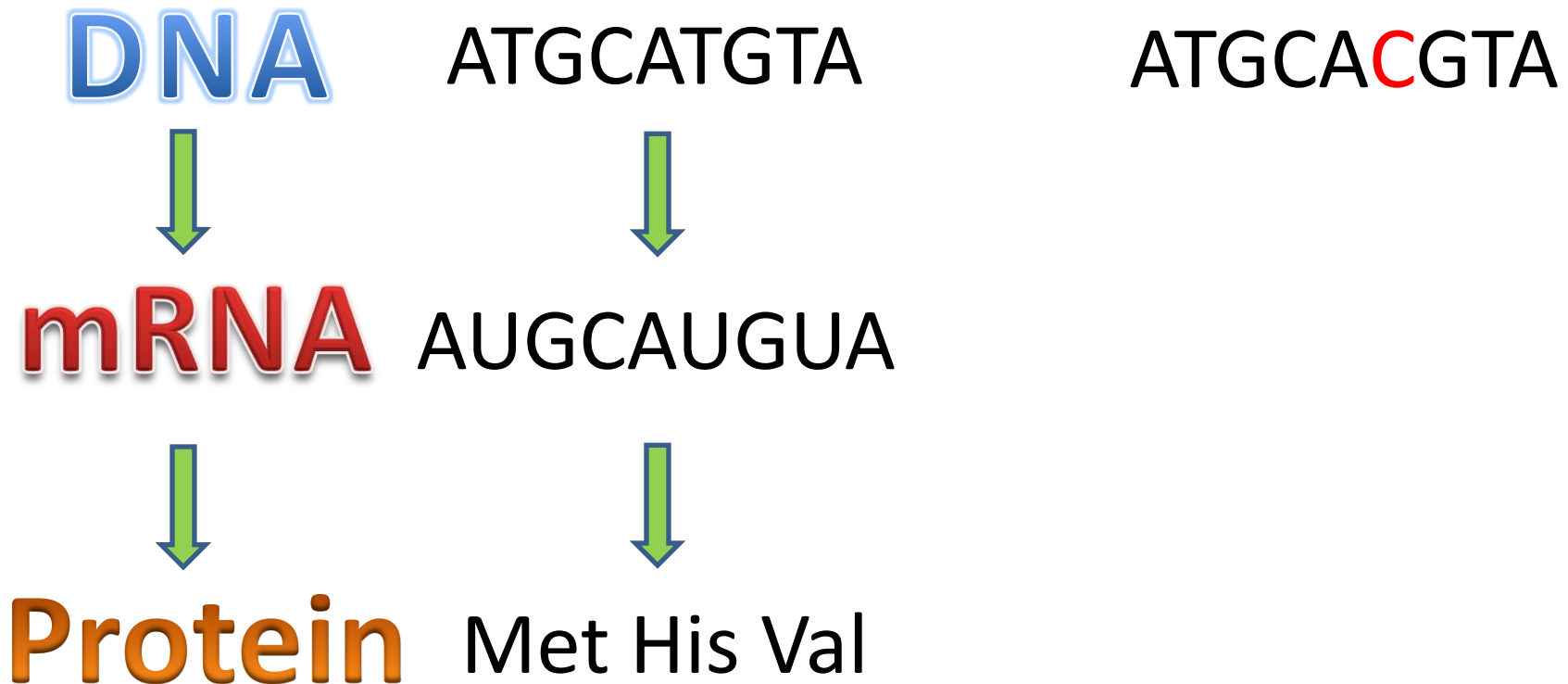
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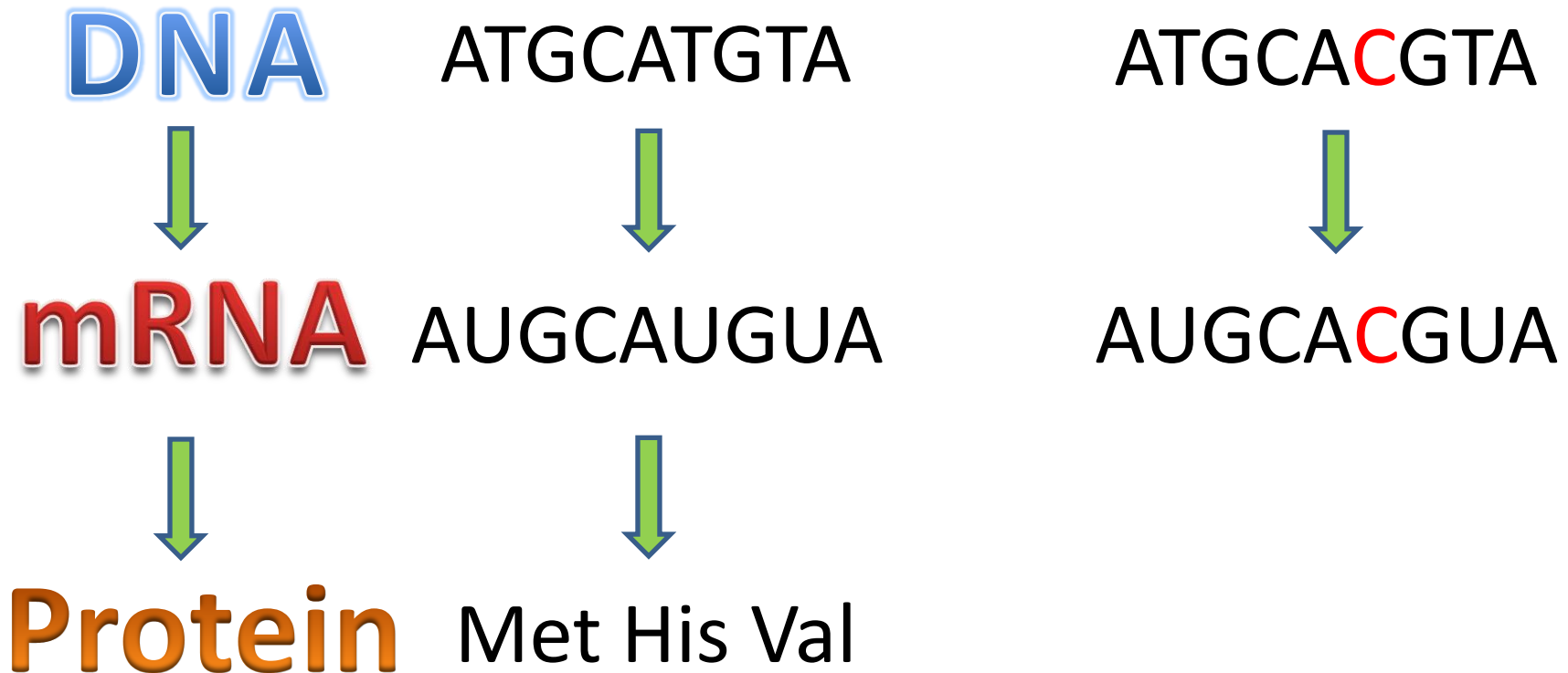
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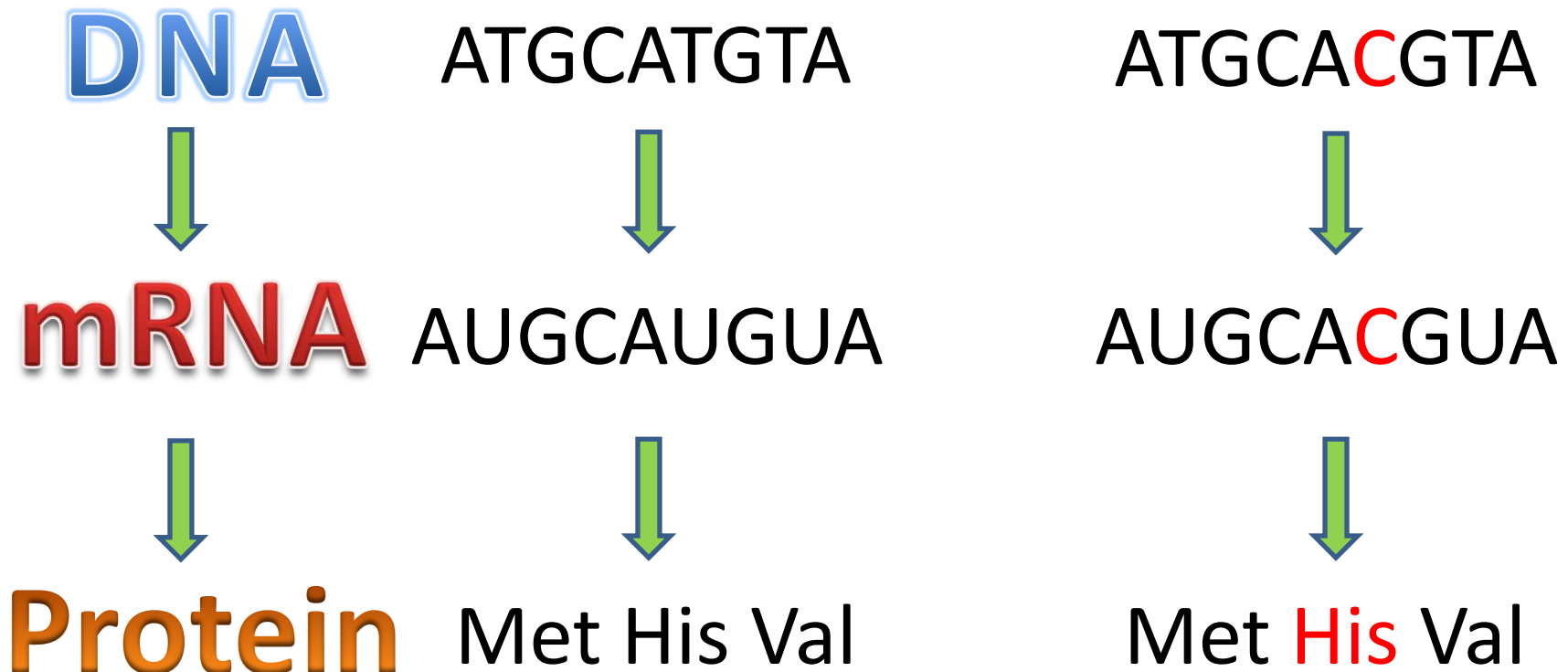
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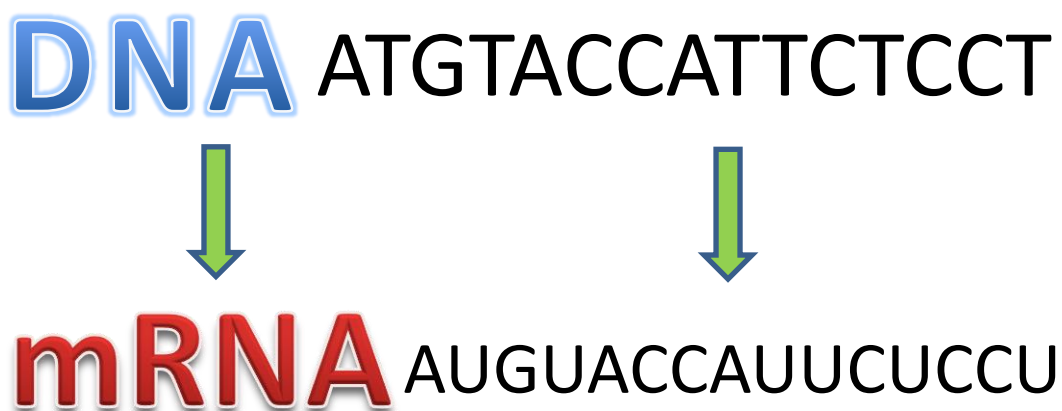
Point Mutation – Non-Sense

- **Non-Sense mutation** – results in a premature stop codon, or a point-nonsense codon in the transcribed mRNA, and in a truncated, incomplete, and usually nonfunctional protein product.

DNA ATGTACCATTCTCCT

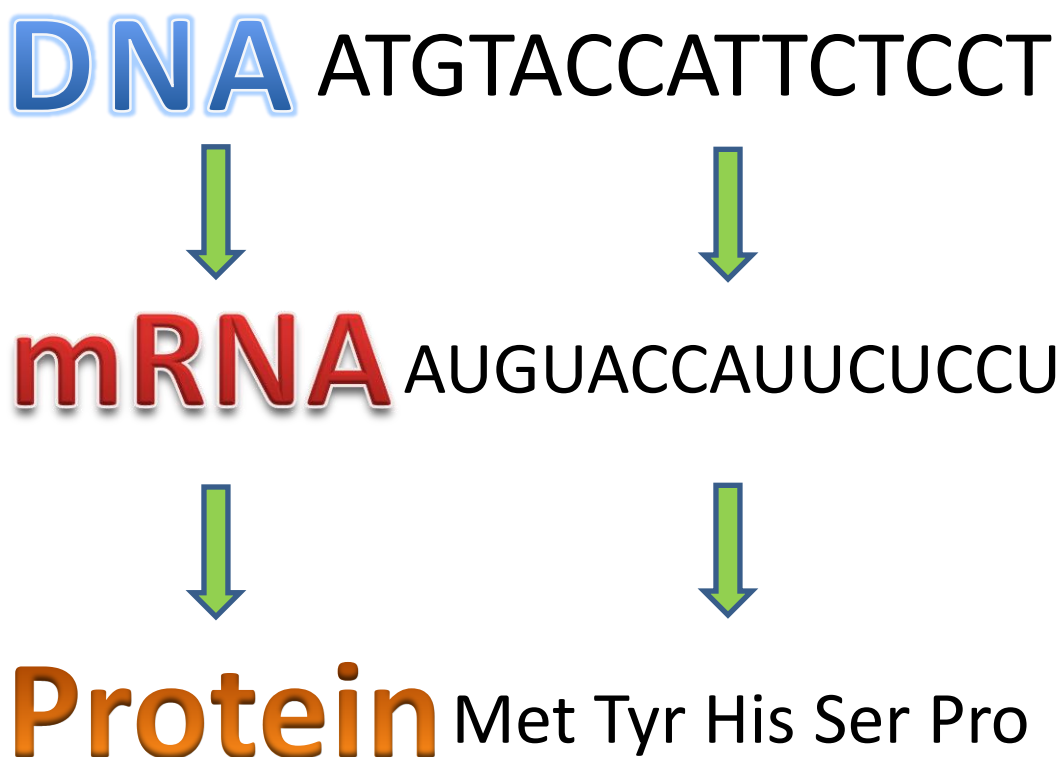
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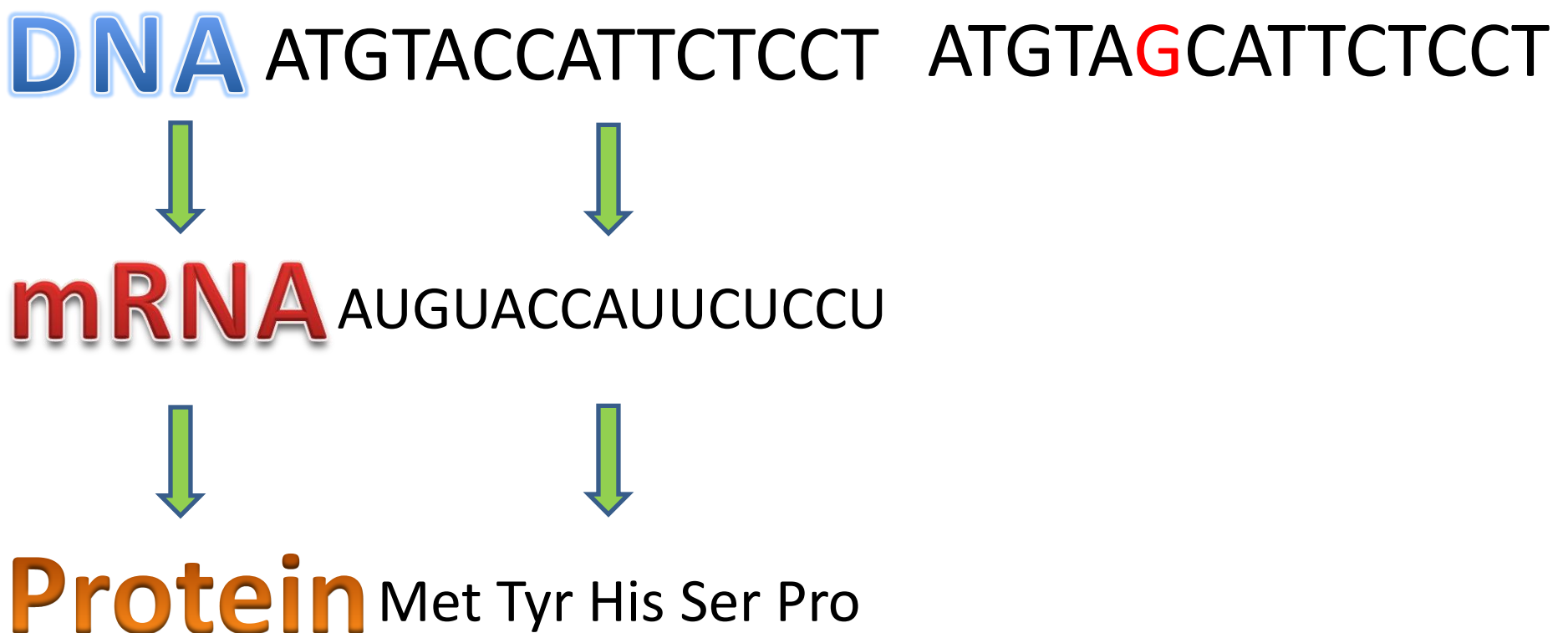
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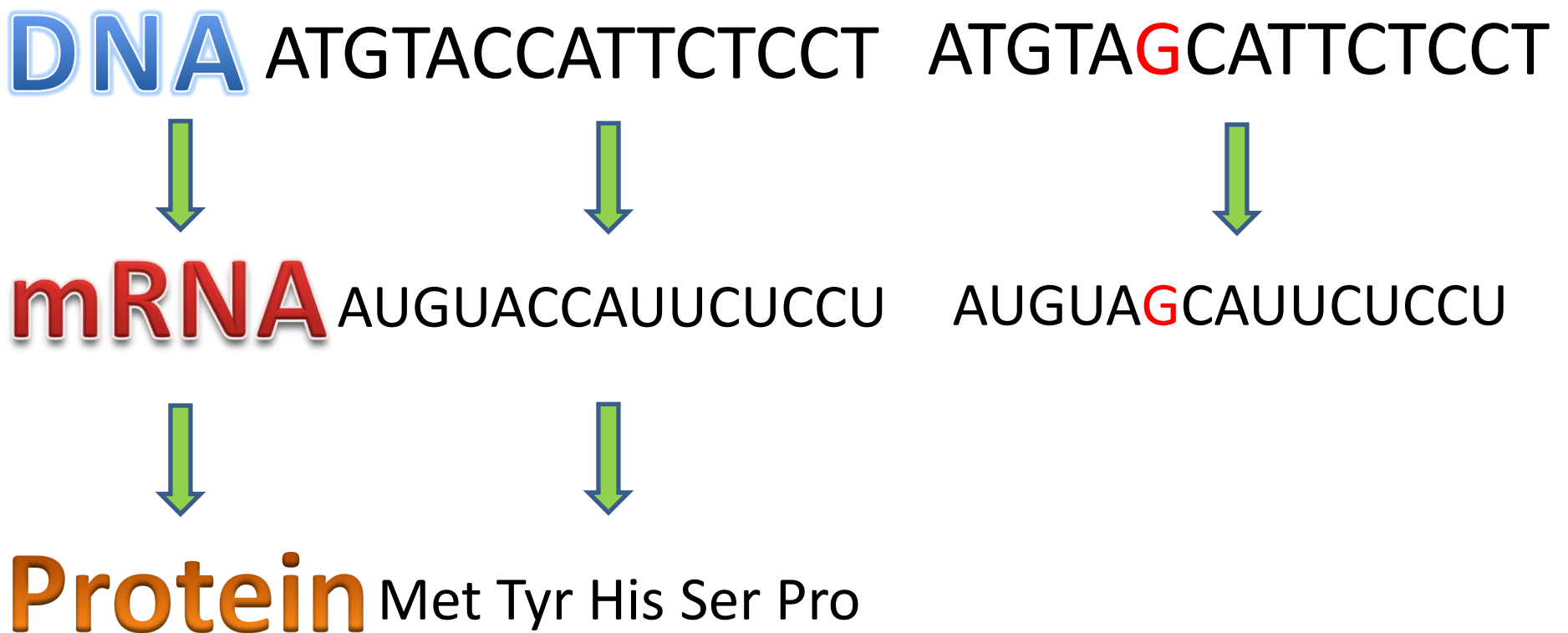
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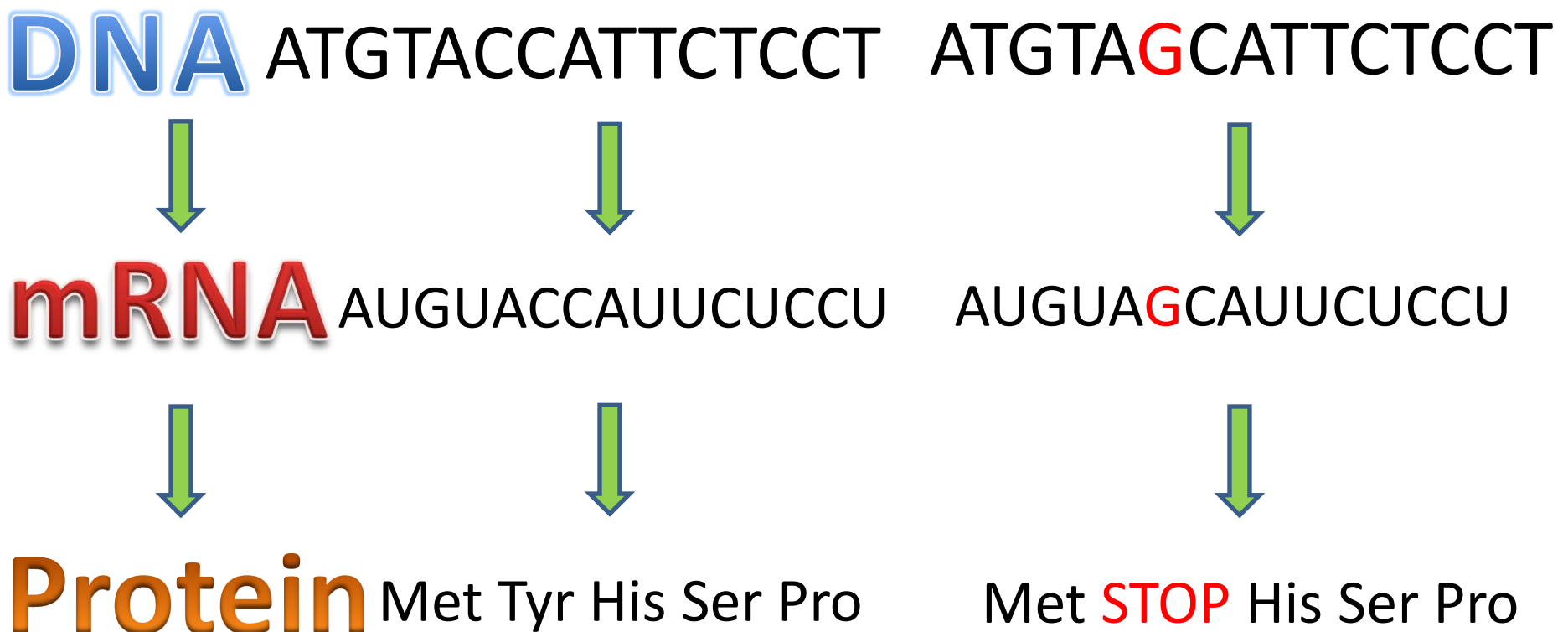
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Protein Met Tyr His Ser Pro

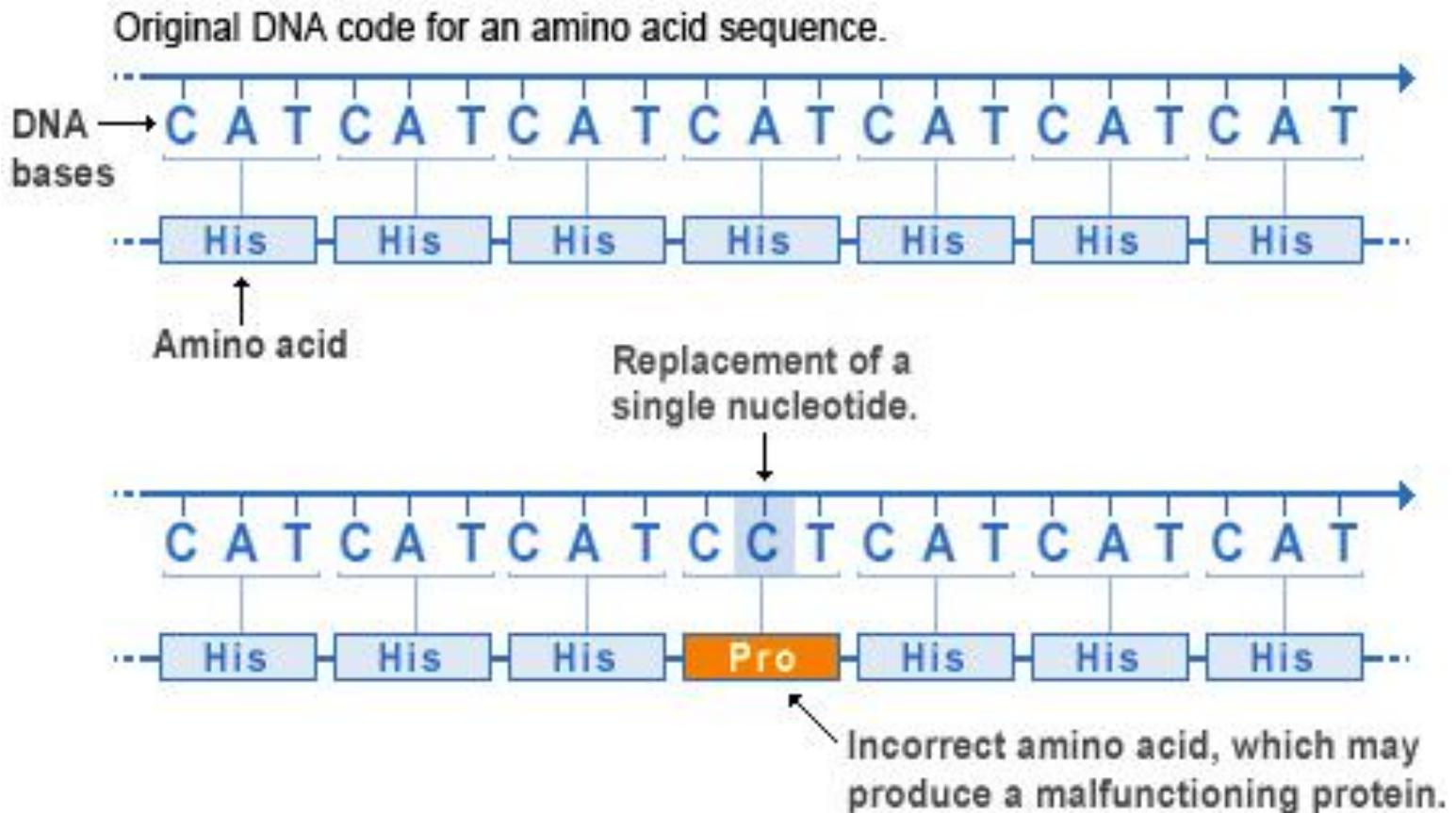
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Point Mutation - Missense

- **Missense mutation** – in which a single nucleotide change results in a codon that codes for a different amino acid.



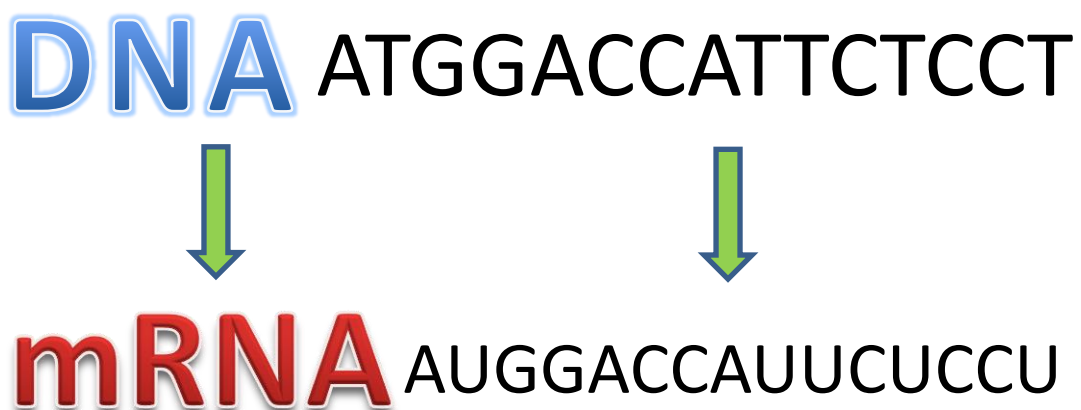
Point Mutation – Missense

- **Missense (conservative) mutation** – A conservative replacement is an amino acid replacement in a protein that changes a given amino acid to a different amino acid with similar biochemical properties

DNA ATGGACCATTCTCCT

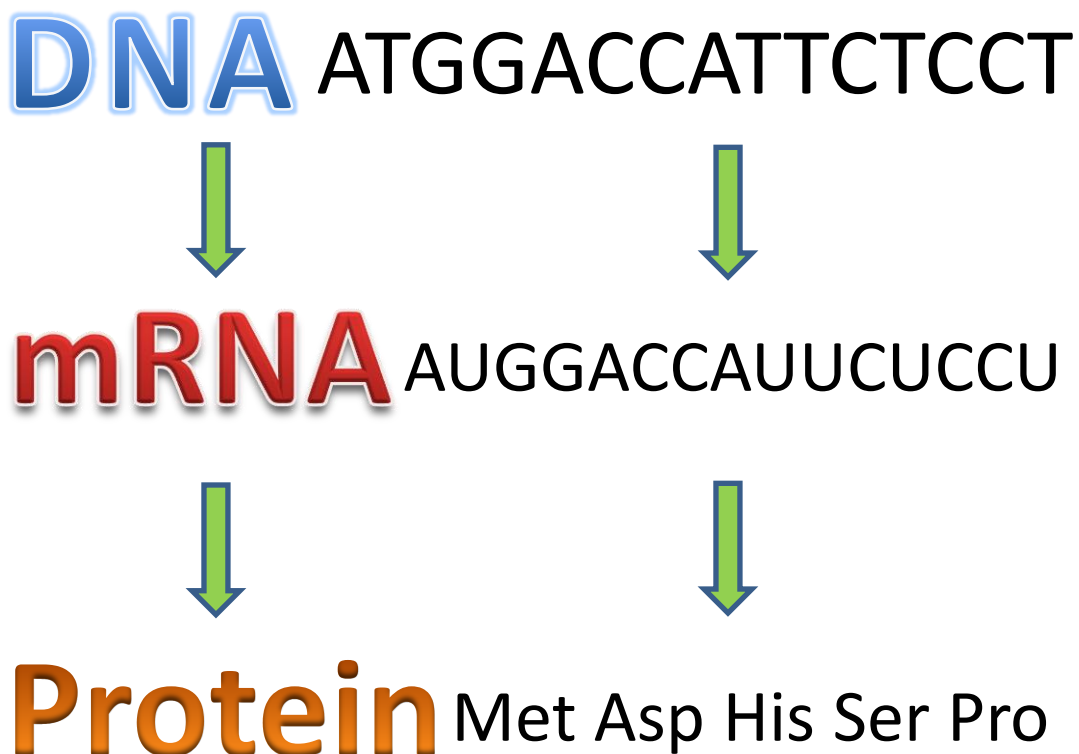
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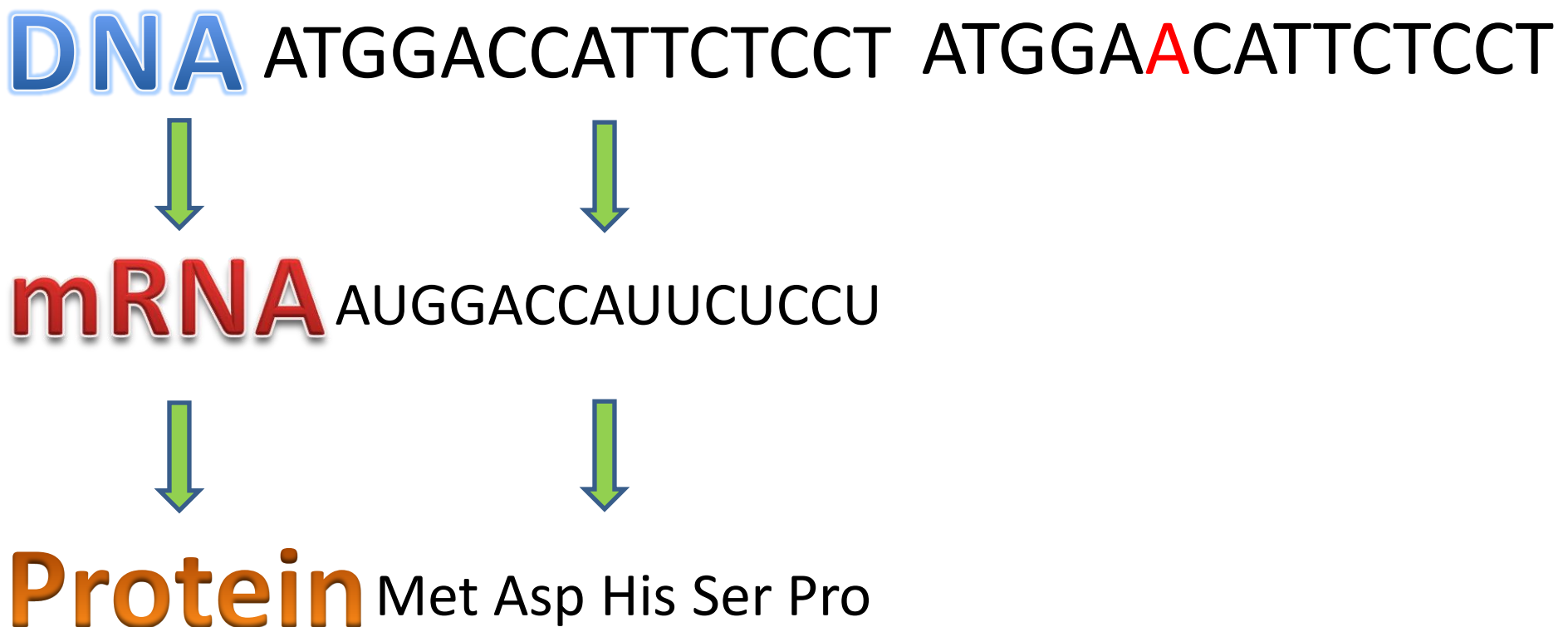
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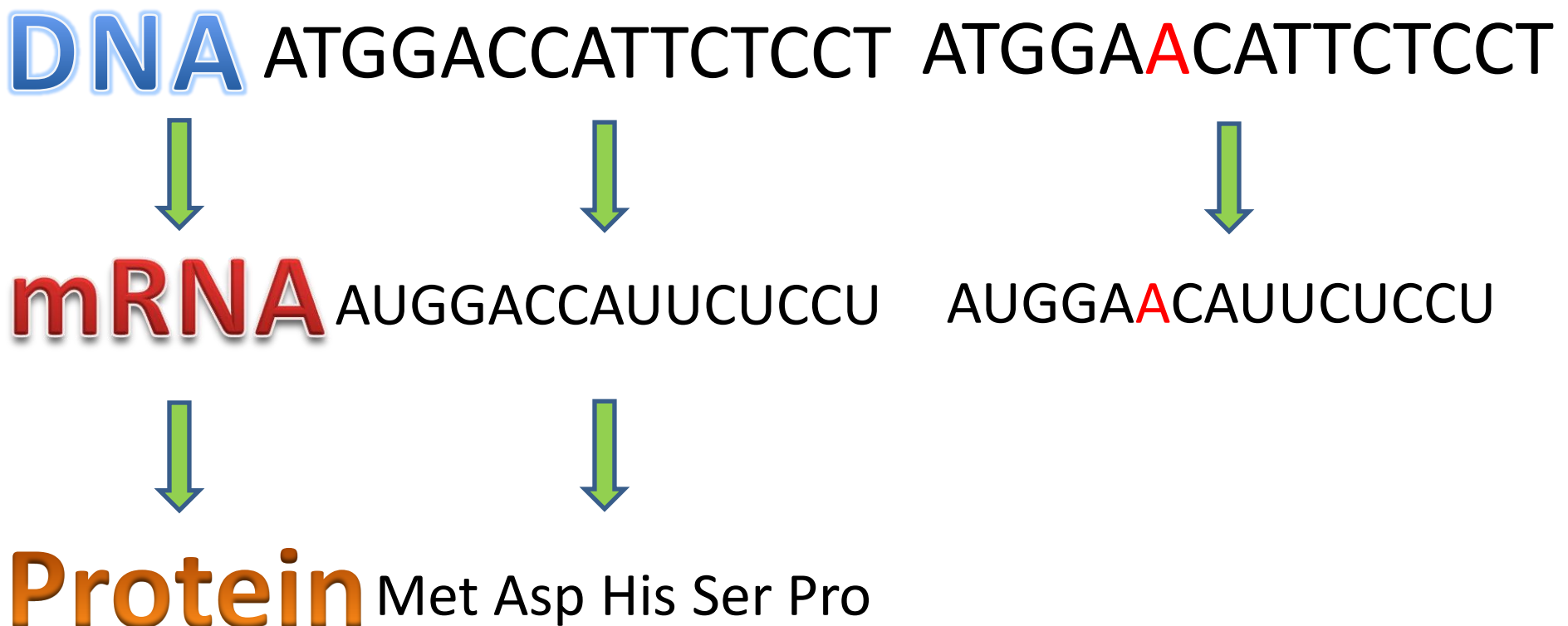
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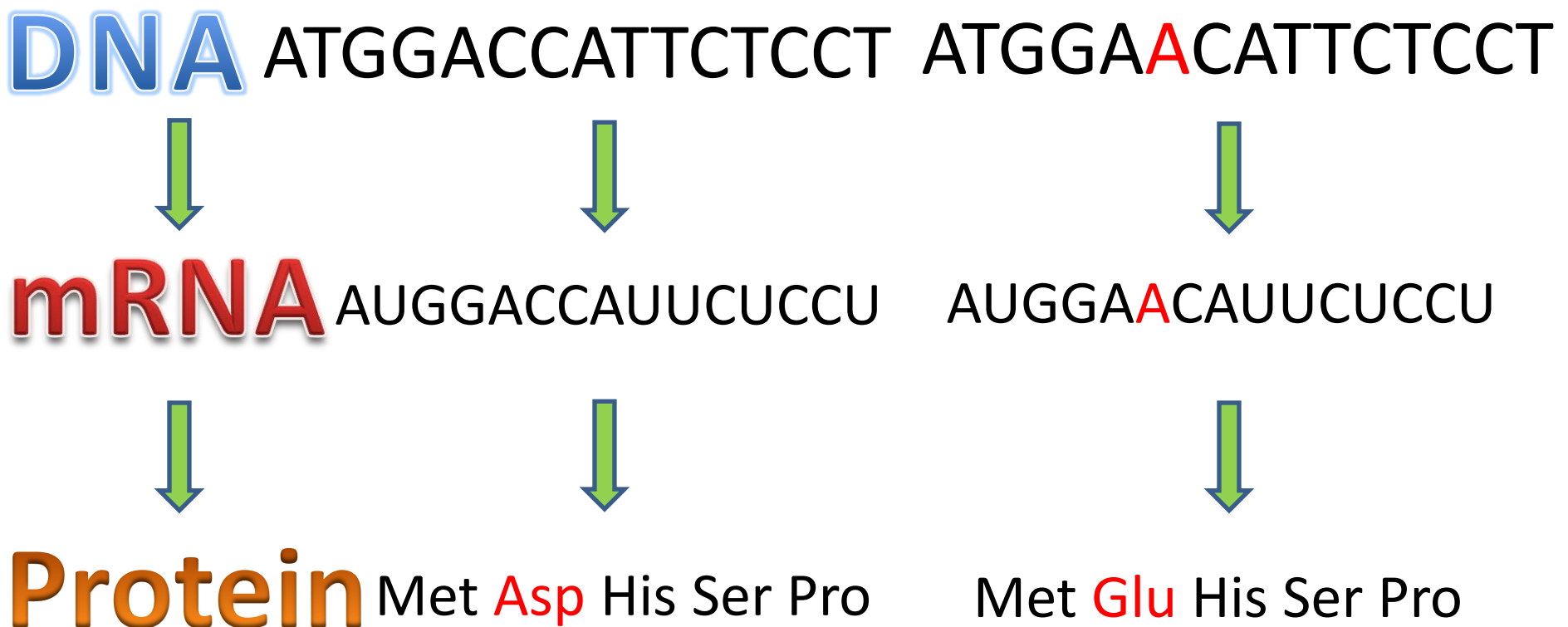
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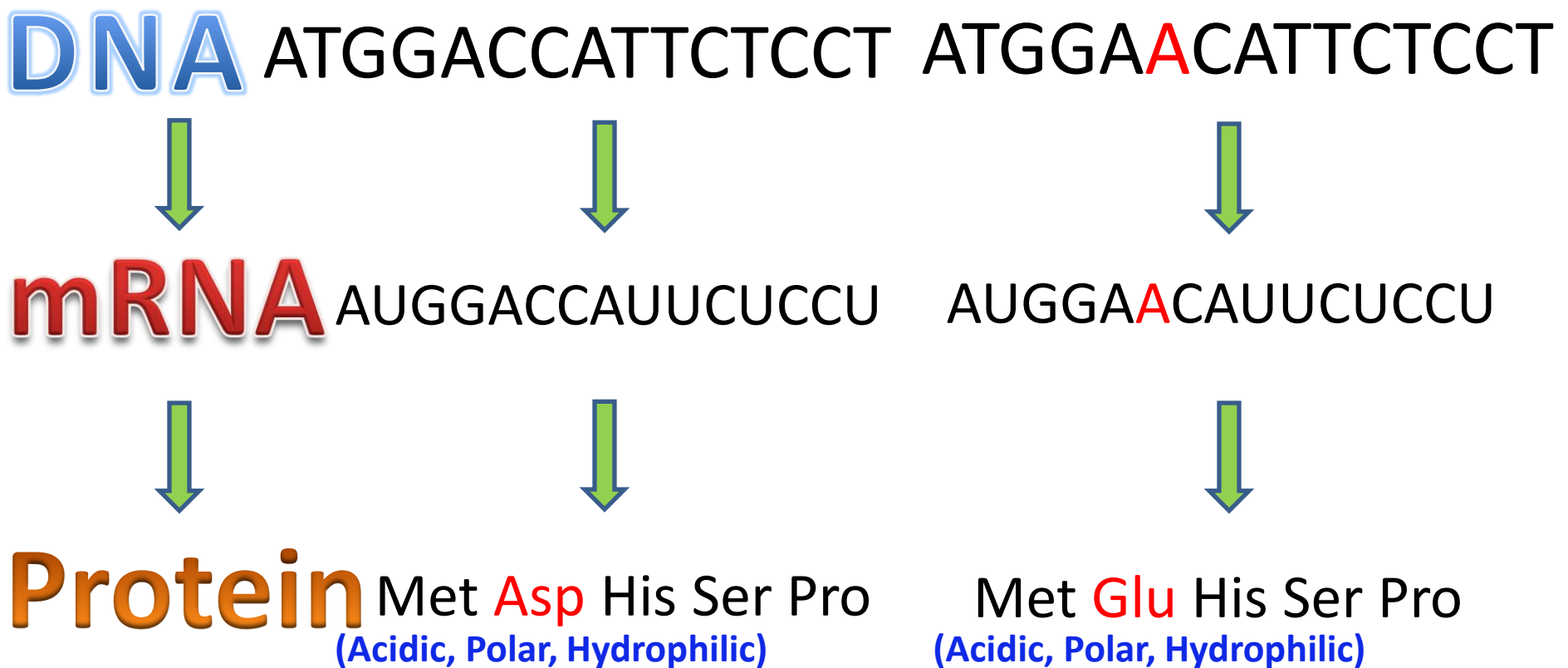
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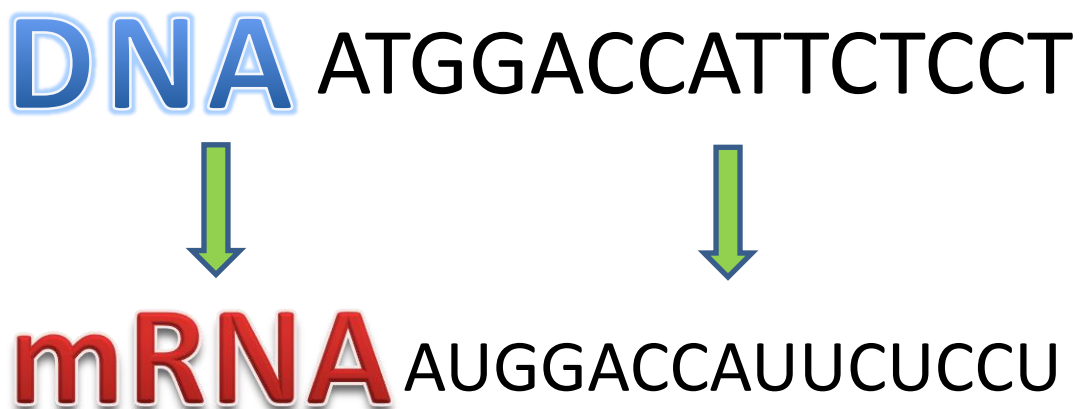
Point Mutation – Missense

- **Missense (non-conservative) mutation** – changes a given amino acid to a different amino acid that has different biochemical properties. The protein may lose its function, which can result in a disease in the organism

DNA ATGGACCATTCTCCT

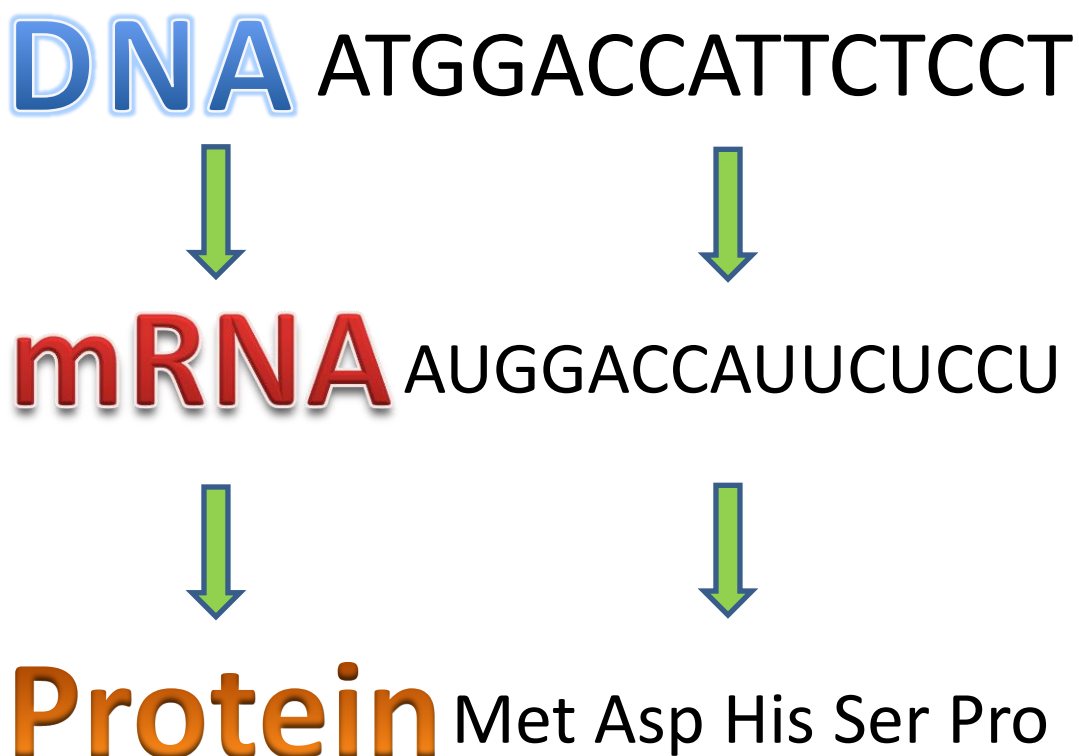
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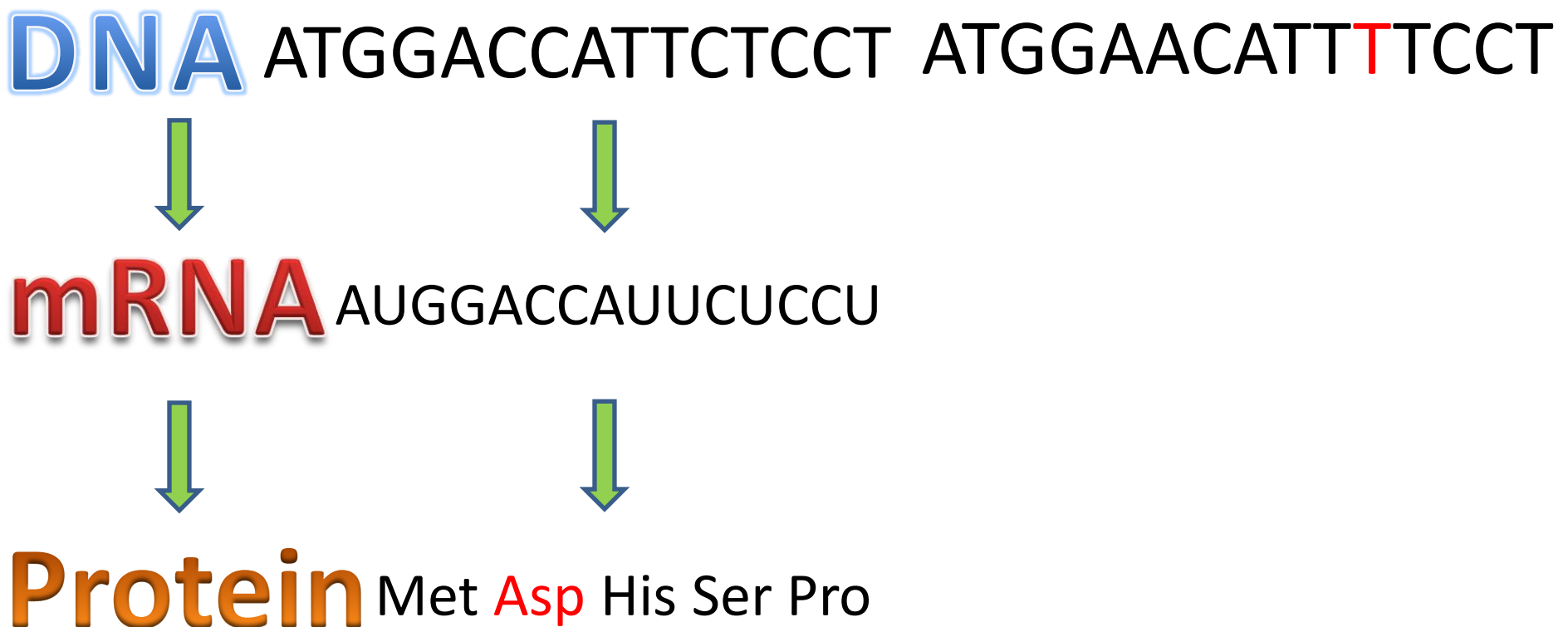
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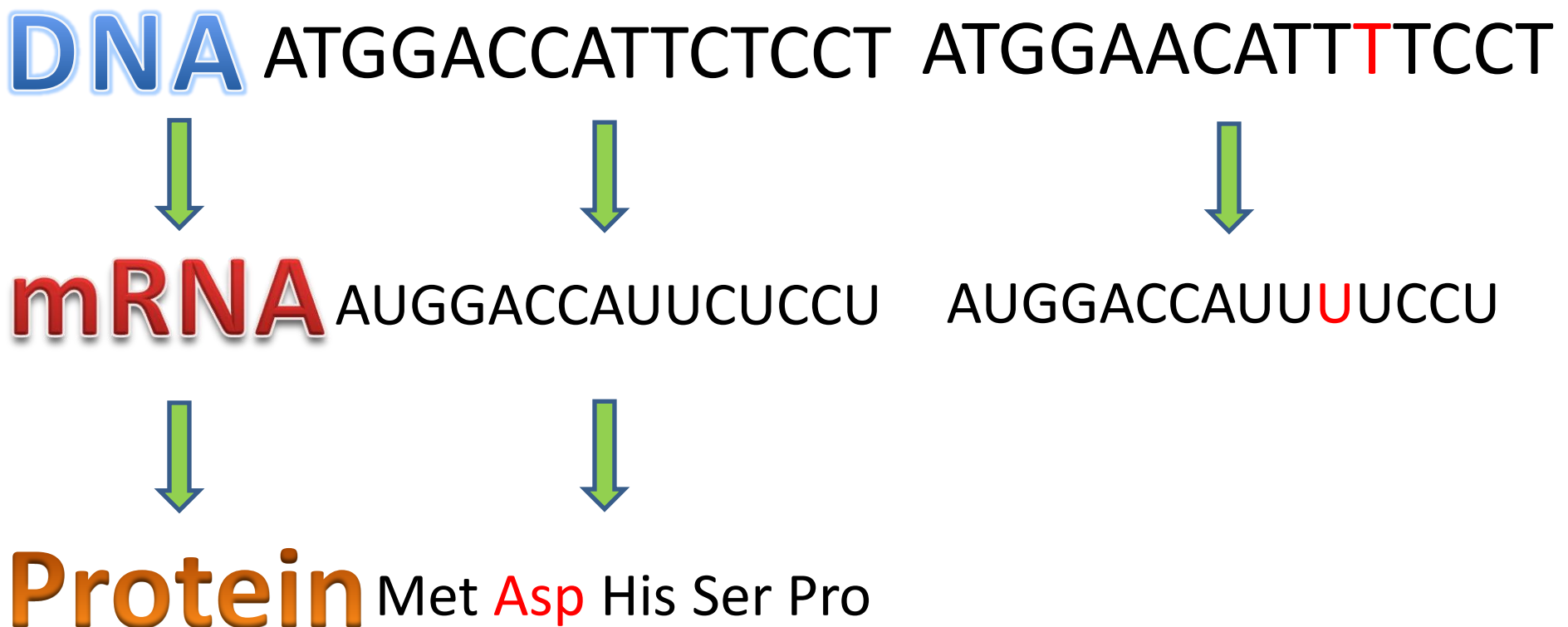
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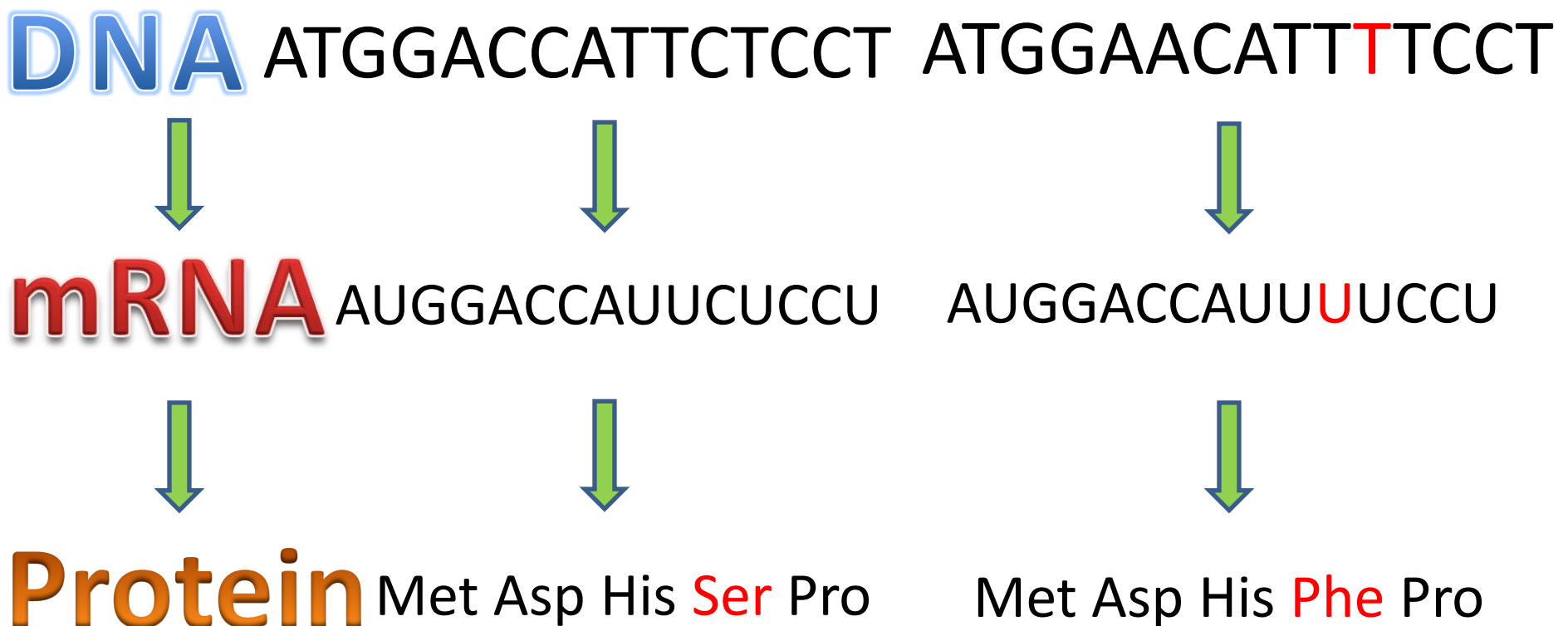
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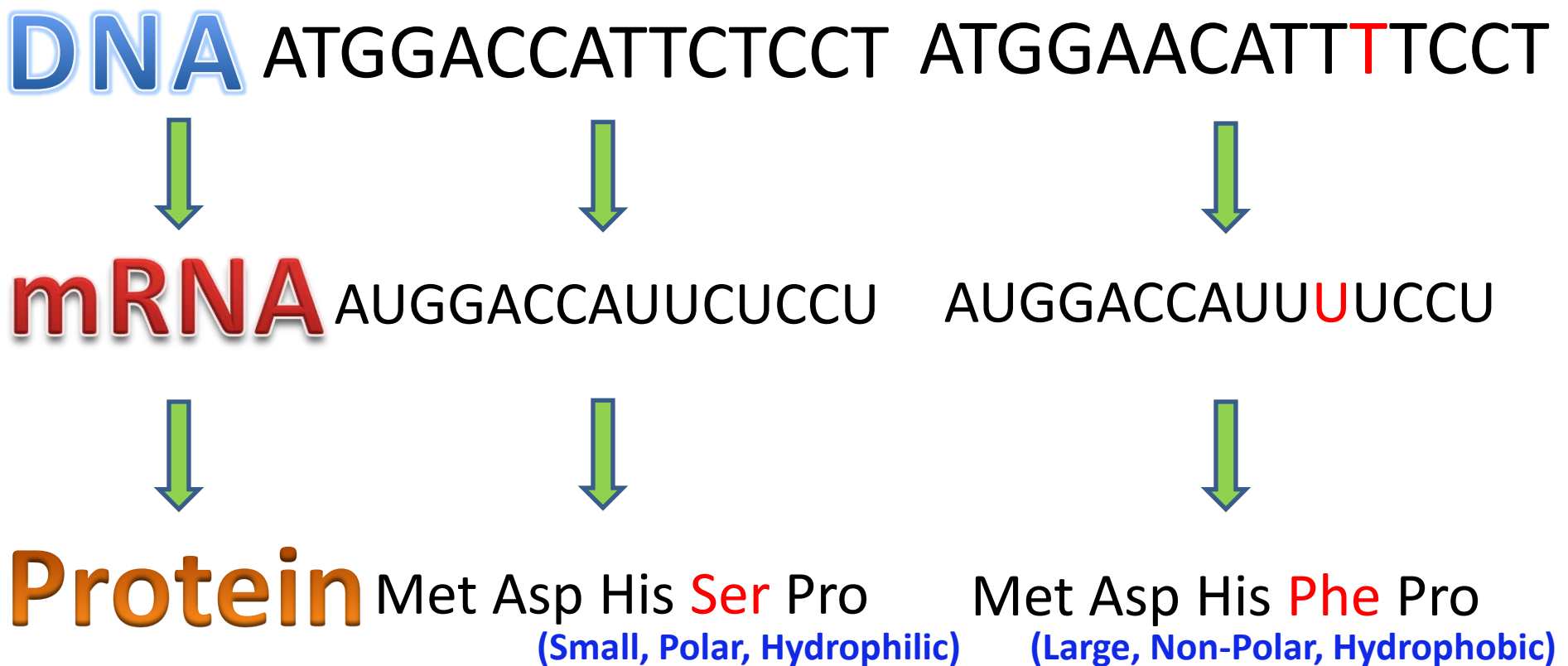
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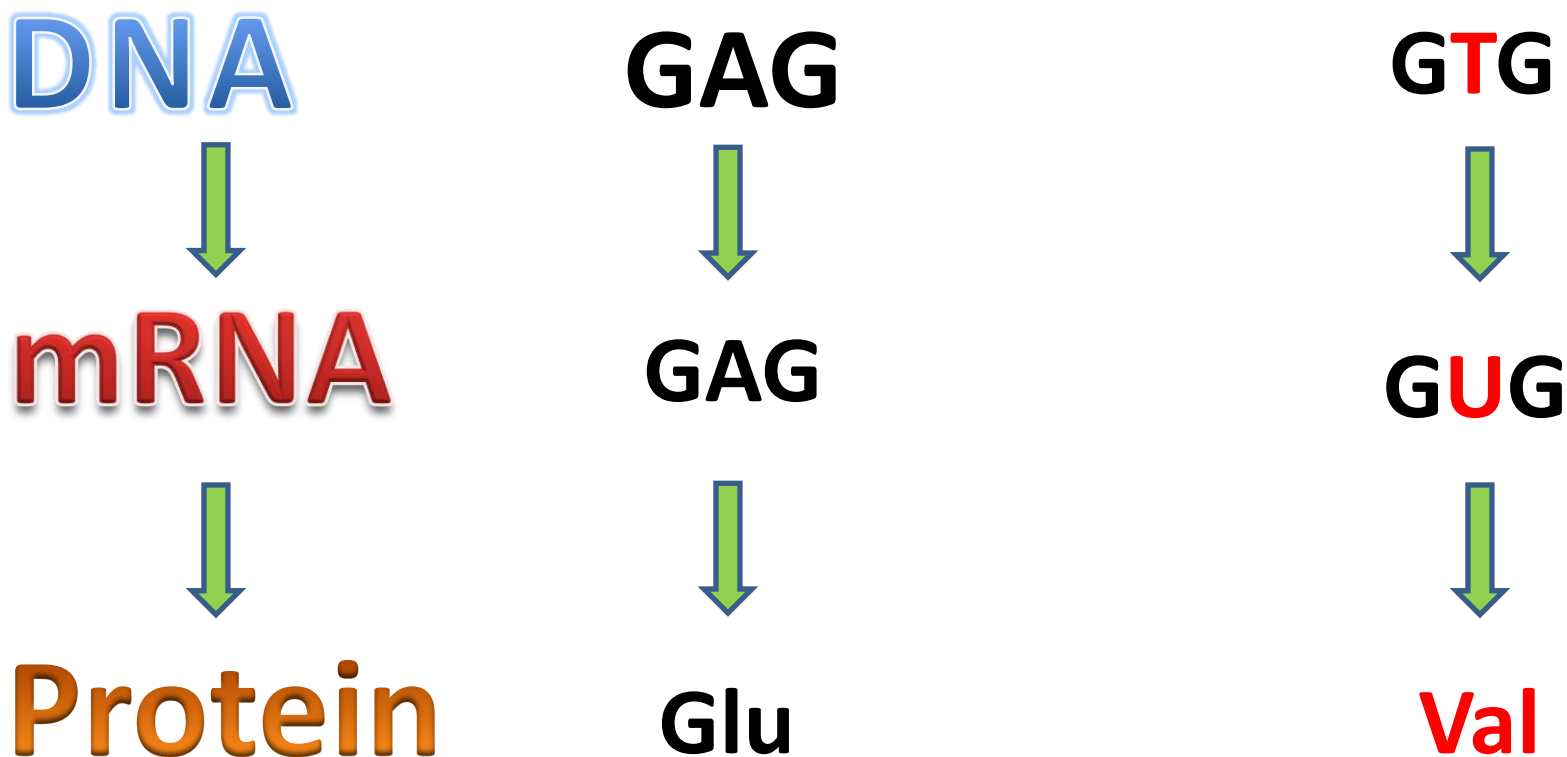
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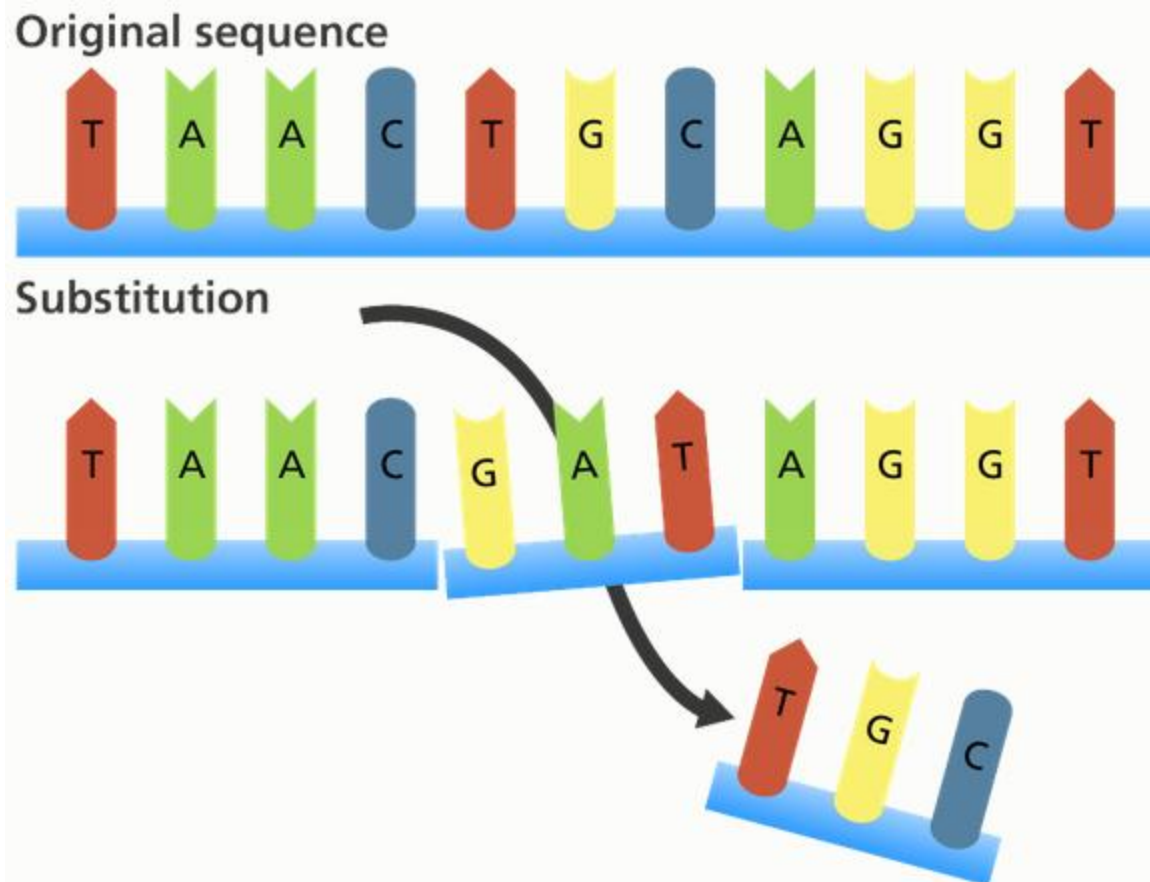
Missense - (Non-Conservative)

- Sickle cell disease** is a group of disorders that affects hemoglobin, the molecule in red blood cells that delivers oxygen to cells throughout the body. In this disease, hemoglobin (Hb) is mutated into a less active form hemoglobin S (Hbs).



Substitution - Mutation

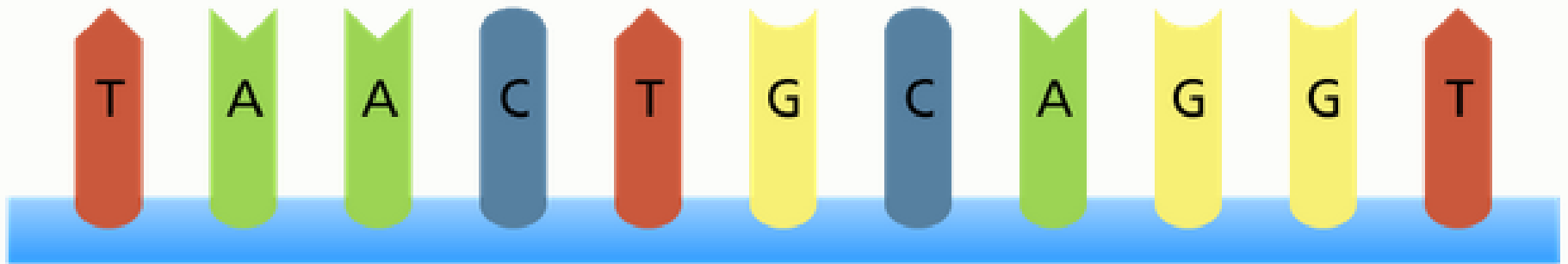
- **Substitution** – when one or more bases in the sequence is replaced by the same number of bases (for example, a cytosine substituted for an adenine)



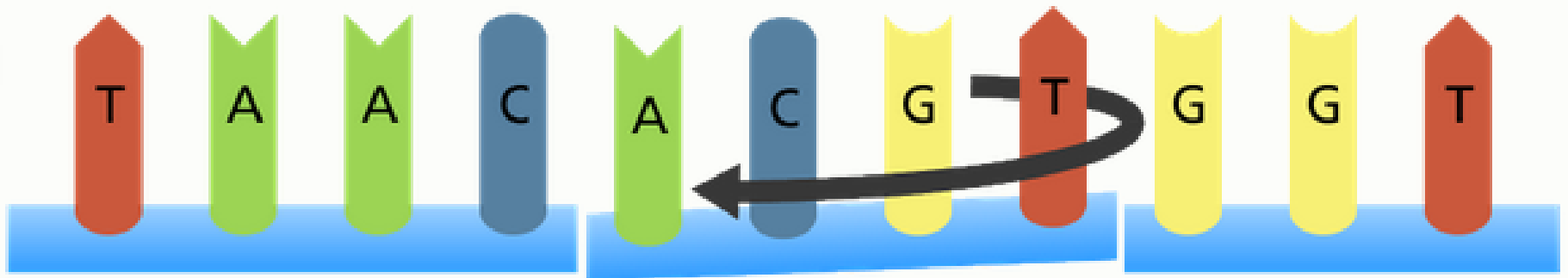
Inversion - Mutation

- **Inversion** – when a segment of a chromosome is reversed end to end.

Original sequence



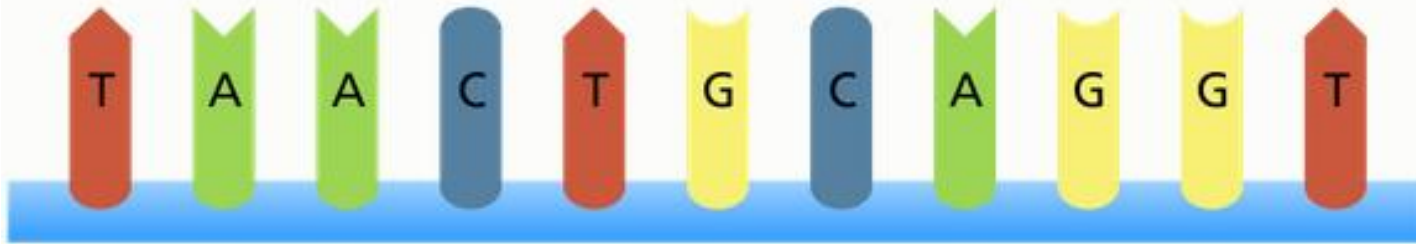
Inversion



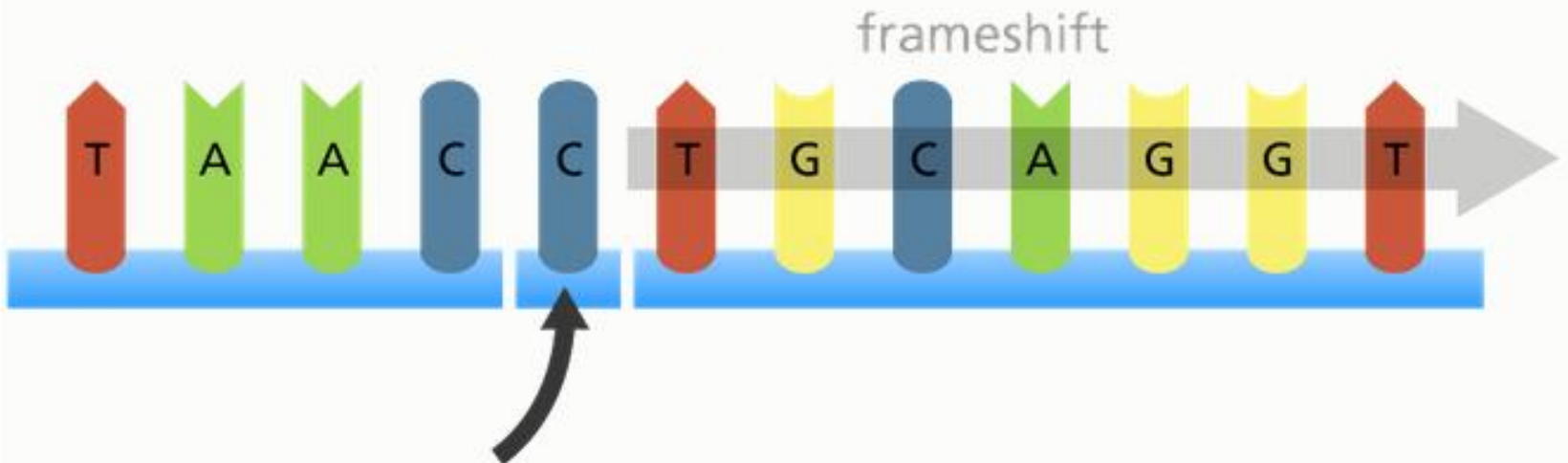
Insertion – Frame Shift Mutations

- **Insertion** – when a base is added to the sequence.

Original sequence

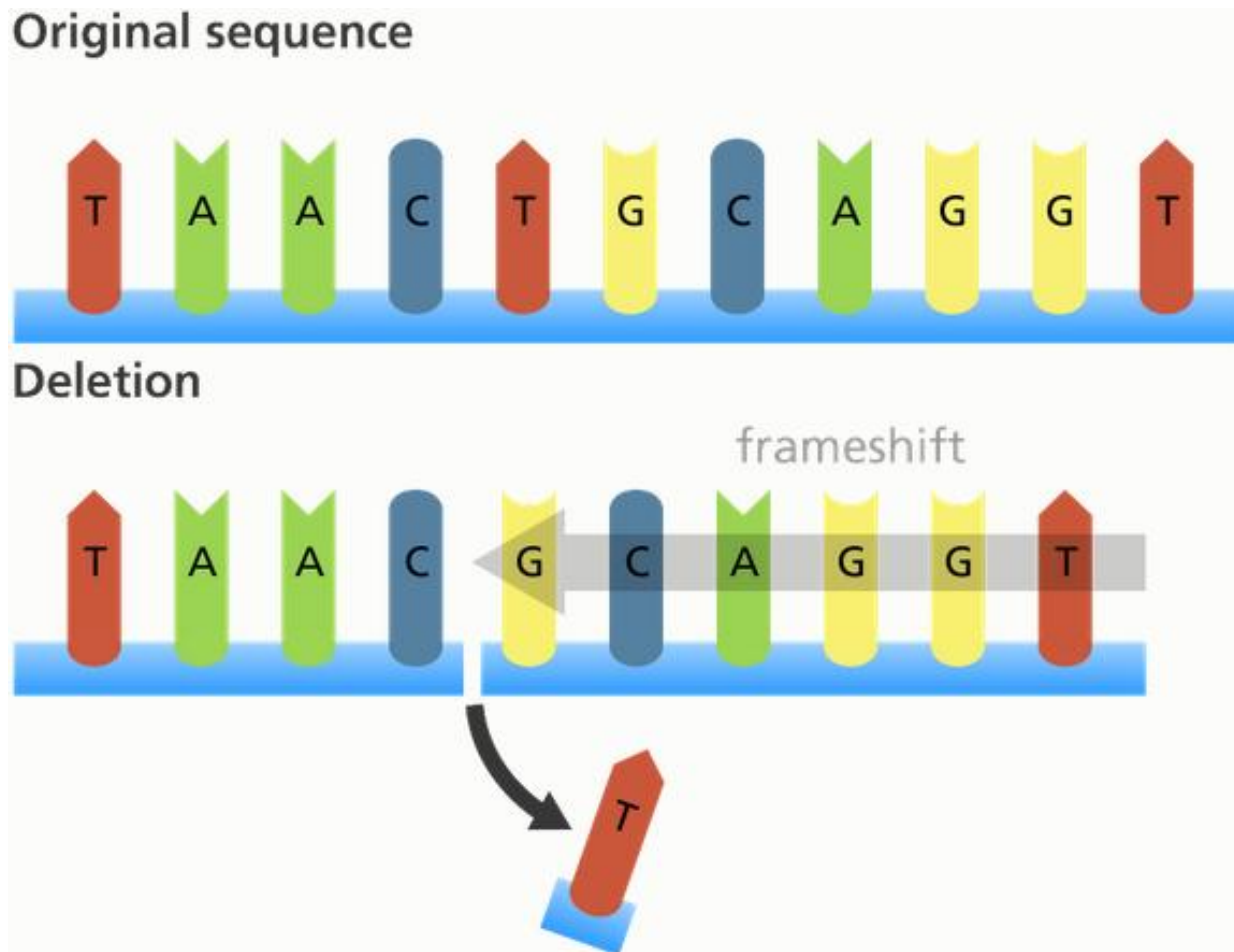


Insertion



Deletion – Frame Shift Mutations

- **Deletion** – when a base is deleted from the sequence.



How do we inherit mutations?

- Each of our genes is a copy from either our mother or our father.
- If there is a mutation in one of these genes, this can be passed on from parent to child along with the rest of their genes.
- Even though mutation is common, inherited diseases are relatively rare.
- This is because inherited diseases are often recessive, which means that a person must have two copies of the mutated gene to get the disease.